

Family Compass Ecosystem

Low-Latency Real-Time Private Positioning Network

PREPARED FOR

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Primary Stakeholder & Lead Client Sponsor

PREPARED BY

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STATUS

Formal Proposal & SOW

1. EXECUTIVE SUMMARY

This Statement of Work (SOW) outlines the strategic objectives, technical architecture, and development timeline for the Family Compass Ecosystem. Family Compass is a private, real-time tracking network engineered to provide immediate, visual reassurance for closed family circles. By deploying a dedicated native mobile environment driven by low-latency event channels, the system ensures that circles of trust remain securely connected without compromising user privacy. The core philosophy of this platform dictates strict data privacy: no public profiles, no stranger access, and no intrusive advertising.

2. OBJECTIVES & STRATEGIC RESOLUTIONS

The system is designed specifically to resolve daily anxieties related to family coordination and safety through actionable, real-time technical solutions.

IDENTIFIED CHALLENGE	TECHNICAL RESOLUTION
Coordinating in Crowds: Difficulty maintaining group proximity in crowded environments.	Interactive Live Map: Real-time avatar pinning with instant spatial orientation and background updates.
The Blackout Score: Uncertainty regarding sudden communication drops.	Real-Time Power Diagnostic: Direct visibility of live device battery percentages and status.
Status Verification: Constant manual communication required to confirm safe transit.	Live Movement Telemetry: Automated streaming of moving/stationary states and exact speed metrics.
Unprotected Geodata: Third-party monitoring and unauthorized access to location history.	Military-Grade Privacy: Secure, invite-only access ensuring geographic data is strictly isolated to authorized circle members.

3. TECHNICAL APPROACH

The software architecture integrates direct hardware telemetry with low-latency background socket synchronization and secure digital authentication. To adhere to strict privacy requirements, zero telemetry analysis is performed server-side. Tracked data is utilized solely for active session broadcasting, guaranteeing absolute client privacy and minimizing systemic vulnerabilities.

4. SYSTEM DELIVERABLES

- **Live GPS Map Integration:** Custom vector map view rendering interactive circles of avatar positions. Leverages responsive WebSocket pipes for continuous background spatial movement updates.
- **Private Telemetry & Status:** Proactive live battery percentage streaming, travel speed tracking (km/h), and motion state detection (driving, walking, idle) optimized for minimal hardware power consumption.
- **Zero-Data Mining Architecture:** Strict exclusion of third-party tracking SDKs. All tracking logs are stored in on-device SQLite caches. Server transit logs are scrubbed dynamically upon WebSocket termination.
- **Admin & Circle Control:** Robust administrative dashboard enabling designated circle creators to revoke invite-codes, eject participants, and dynamically purge live-map histories.

5. DEVELOPMENT TIMELINE & MILESTONES

The project is structured into four primary phases to ensure systematic development, rigorous testing, and on-time deployment.

Timeline Provision: The target development timelines specified below represent the baseline operational schedule. By mutual verbal or written agreement between both parties, acceleration measures may be applied to conclude individual phases or the entire project ahead of the stated timeline.

PHASE	CORE DELIVERABLES	TARGET TIMELINE
1: Foundation	Database schemas, secure local sign-in routines, and core API layer configuration.	Weeks 1 to 2
2: Telemetry Engine	Live GPS WebSocket connection and low-power background synchronization protocols.	Weeks 3 to 4
3: Interactive UI	Responsive map views, circles controller dashboard, and battery notification alert integration.	Weeks 5 to 6
4: Launch & QA	Rigorous security testing, payload optimization, and staging launch.	Weeks 7 to 8

6. INVESTMENT SUMMARY & MILESTONE PAYMENTS

Payments will be structured progressively and invoiced upon the successful delivery and client approval of each milestone. In alignment with infrastructure requirements, ongoing backend server hosting, cloud database storage, and related maintenance costs are excluded from this development quote and remain the direct responsibility of the client.

MILESTONE INVOICE	TRIGGER EVENT	%	AMOUNT (ETB)
Milestone 1 Foundation	Completion of database schemas, secure sign-in, and API layer.	30%	45,000
Milestone 2 Telemetry Engine	Completion of WebSocket connections and background sync engine.	30%	45,000
Milestone 3 Interactive UI	Completion of responsive map, circle dashboard, and alerts.	25%	37,500
Milestone 4 Launch	Completion of security testing, optimization, and staging deployment.	15%	22,500
TOTAL PROJECT INVESTMENT		100%	ETB 150,000

7. THIRD-PARTY PUBLISHER FEES & COSTS

Upon development completion, the application will be submitted to the official app stores. The administrative process of submission is included at no additional development cost. However, mandatory developer platform fees are paid directly to the respective publishers and are **not paid to the contractor**.

PLATFORM	STANDARD FEE	PAYEE / RECIPIENT
Google Play Store	Approximately \$25 to \$35 USD (One-time fee)	Google
Apple App Store	\$99 USD (Annual fee)	Apple

8. TESTING & LAUNCH PROCESS

Prior to public release, the application will undergo a structured quality assurance protocol to ensure stability and reliability.

- **Targeted User Testing:** 12 diverse users testing the application across multiple operating systems and hardware profiles.
- **14-Day Testing Window:** A dedicated 2-week active testing period to isolate edge cases and patch application logic.
- **Optimization Audits:** Comprehensive performance indexing, battery consumption diagnostics, and interface quality checks.
- **Store Submission & Launch:** Official package delivery and deployment to both the Google Play Store and Apple App Store. (Note: Final approval timelines are strictly subject to Google and Apple review queue fluctuations).

9. TERMS & ASSUMPTIONS

- **Validity:** This proposal and associated pricing remain valid for 30 days from the document date.
- **Payment Terms:** Milestone invoices are strictly due within 7 calendar days of milestone delivery and review.
- **Scope Alterations:** Any features, design changes, or deliverables requested outside of this finalized SOW require a formal change order and will be quoted separately.
- **Intellectual Property:** Full intellectual property rights and raw source code ownership are transferred to the client upon receipt of the final milestone payment.
- **Server Infrastructure:** Consistent with prior technical discussions, ongoing backend server hosting, cloud database storage, and maintenance costs are excluded from this development quote and remain the responsibility of the client.
- **Confidentiality:** The strategies, code, and terms within this document are strictly confidential.

10. ACCEPTANCE & SIGNATURES

By signing below, both parties agree to the scope, milestones, and terms outlined in this Statement of Work, authorizing the immediate commencement of Phase 1 development.

Bisrat Abrham

Senior Full Stack Developer (Contractor)

Date: _____

Kalab Nardos

Lead Stakeholder Sponsor (Client)

Date: _____

11. CONTACT INFORMATION

Bisrat Abrham

Senior Full Stack Developer

Available via standard, established communication channels for project management and technical inquiries.